

Layer	Feature	Technical Description	User/System Benefit	Person Assigned
Sequencing (Go)	FIFO Ordering Engine	Implements a First-In-First-Out queue using Go channels and mutexes to prevent race conditions.	Eliminates "priority gas auctions" where users overpay to jump the line.	Vidhan
	Sequence Commitment	Generates a cryptographic hash (commitment) of the order metadata as soon as it hits the server.	Provides "Proof of Reception" so the sequencer cannot rearrange orders later.	Prasham
	High-Concurrency RPC	Uses a Gin or gRPC framework to handle thousands of incoming TCP connections simultaneously.	Ensures the system stays responsive during high-market volatility.	Drumil
Privacy (C++)	NaCl Sealed Boxes	Utilizes libsodium for public-key authenticated encryption. Transactions are "sealed" by the user's key.	Hides trade direction (Buy/Sell) and volume from prying eyes (MEV bots).	Dhairya
	Low-Latency Decryption	C++ optimized routines for batch-decrypting orders once they reach the "Execution" stage.	Maintains high throughput; privacy doesn't become a bottleneck for speed.	Drumil
	Encrypted Mempool	A temporary storage area where all pending transactions stay encrypted until the sequencer triggers the "reveal."	Prevents "sandwich attacks" because bots cannot calculate the impact of the trade.	Devarsh
Settlement (Solidity)	On-Chain Verification	A smart contract that checks the validity of the decrypted data against the initial Go-layer commitment.	Ensures the Sequencer didn't tamper with the transaction data during the process.	Prasham
	Mock DEX Engine	A Solidity-based Automated Market Maker (AMM) that simulates liquidity pools (like Uniswap).	Allows for risk-free testing of trade execution, slippage, and fee distribution.	Vidhan
	Atomic Settlement	Logic that ensures the swap and the fee payment happen in a single, irreversible transaction.	Prevents partial trades or "failed" states that could lead to loss of funds.	Dhairya
Simulation (Python)	DEX Modeling	A script that acts as a "Liquidity Provider" and "Trader" to stress-test the system.	Helps you visualize how the blockchain handles heavy traffic before deployment.	Devarsh
	Latency Analytics	Tools to measure the time taken from "Order Sent" to "Order Settled."	Provides data for your project report on the efficiency of your architecture.	Prasham